

PORTFOLIO CASE STUDY

Sales & Profitability Intelligence

From raw transactional data to SQL data quality, Python analysis and an interactive Power BI decision dashboard.

ANALYTICAL PERIOD

January 2021 – December 2025

PRIMARY OBJECTIVE

Understand revenue, profit and margin leakage

TECH STACK

SQL Server · Python · Power BI

DASHBOARD

4-page interactive Power BI report

Executive question

Where is the business creating revenue without creating enough profit?

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This case study demonstrates an end-to-end analytics workflow. The underlying dataset appears synthetic; the findings therefore demonstrate analytical capability and should be validated on real business data before being used for decisions.

1. Executive Summary

Between 2021 and 2025 the business realised **144.2M in revenue** and **59.8M in profit (41.4% margin)** from 113,559 completed orders placed by 24,748 customers. Headline performance is healthy, but profit is held back by four structural issues.

144.2M Revenue	59.8M Profit	41.4% Profit margin
113,559 Completed orders	24,748 Active customers	1,270 Average order value

- 1. Peak-season revenue does not turn into profit.** November and December revenue is 65% and 76% above a typical month and makes up 25% of five-year revenue, yet profit in those months is only 7% and 15% higher. Margin drops from 46–47% in regular months to 29.5–29.8%. The cause is confirmed: the average discount is 31.7% in November–December against 11.1% in regular months, and 98.9% of all loss-making orders are placed in those two months. Of the 15.2M of extra revenue earned in peak months, only 1.1M became profit (7 cents per unit of revenue, against 45 cents normally).
- 2. Discounting is the main margin lever and it is under-controlled.** 84.8% of orders carry a discount of up to 30% and earn a 44.9% margin. The 13.8% of orders discounted 30–50% earn 26.2%, and the 1.4% discounted above 50% earn 1.8%. All 1,229 loss-making completed orders (total loss 191K) sit above the 30% discount line, and 45% of orders in the 50–70% band lose money. The direct loss is small (0.3% of profit); the real cost is margin dilution, estimated at about 5.2M (8.6% of profit).
- 3. Growth is concentrated in the lowest-margin categories.** Electronics produces 21.4% of revenue at a 32.6% margin, 21.5 points below Grocery (54.1%). The three largest categories (Electronics, Jewelry, Home Appliances) generate 47.6% of revenue but only 41.7% of profit.
- 4. Customer segments do not separate value, and 18.6% of booked order value never becomes revenue.** Consumer, Business, Premium and VIP customers all spend about 6.3K and place about 4.6 orders each, so the labels are not behaviourally meaningful. Of 177.1M booked across all statuses, 19.7M is cancelled or still pending and 13.2M is returned.

#	Priority recommendation	Estimated impact (five-year scenario)
1	Hard 40% discount cap, with approval between 30% and 40%	1.3–2.6M by capturing 25–50% of the 5.2M margin leakage
2	Replace blanket Nov–Dec discounts with targeted bundles	About 0.37M per margin point recovered in peak months
3	Protect Electronics margin (bundles, discount depth)	About 0.33M per margin point
4	Rebuild segmentation with RFM	Enables targeted retention and pricing
5	Investigate cancellations, returns and unpaid completed orders	A 10% cut in returns protects about 1.3M of booked value

Estimates are scenarios, not forecasts, and recommendations 1 and 2 address overlapping margin leakage, so their impacts must not be added together.

2. Objective and Scope

The project answers three business questions: where the business makes and loses money; which factors drive margin (discount, category, customer segment, region, season); and how reliable the reported KPIs are.

Deliverables: a documented SQL data-quality and clean layer, a Python analysis notebook, and a four-page Power BI dashboard (Executive Overview, Product & Category, Customers & Regions, Discount & Loss).

3. Data and Method

Pipeline.

01 SQL SERVER Audit, reconcile, build clean views	02 PYTHON EDA, business analysis, CSV outputs	03 POWER BI Star schema and 4-page dashboard
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The Power BI model is a star schema: dim_Customers, dim_Products and dim_Date, with fact_Orders (order headers), fact_OrderLines (order items) and a dedicated _Measures table.

Table	Grain	Rows
customers	One row per customer	25,000
products	One row per product	1,175
orders	One row per order line (items)	397,569
sales_customer	One row per order (headers)	138,116

KPI	Definition
Revenue	Sum of net_sales for orders with status Completed (header table)
Profit	Sum of profit for Completed orders
Profit margin	Profit / Revenue (always revenue-weighted, never an average of order margins)
AOV	Revenue / number of Completed orders
Discount %	discount_amount / gross_sales
Loss order / loss rate	Completed order with profit below 0 / loss orders divided by Completed orders
Product-level metrics	Calculated from order lines (items) joined to Completed orders

Key design decisions.

- Only Completed orders count as revenue. Returned orders carry a Refunded payment status and Cancelled orders a Failed one.
- Headline KPIs come from order headers; category, product, segment, region and discount-band tables come from order lines. The two sources differ by about 8.7% (section 4), so **tables on a line basis total 156.8M, not 144.2M**, and the two views are never added together.
- Currencies are not converted because average order value is similar in all seven currencies (about 1,238–1,384), which indicates a common scale.
- Raw tables are never modified; cleaning happens in customers_clean and in views.

4. Data Quality Audit

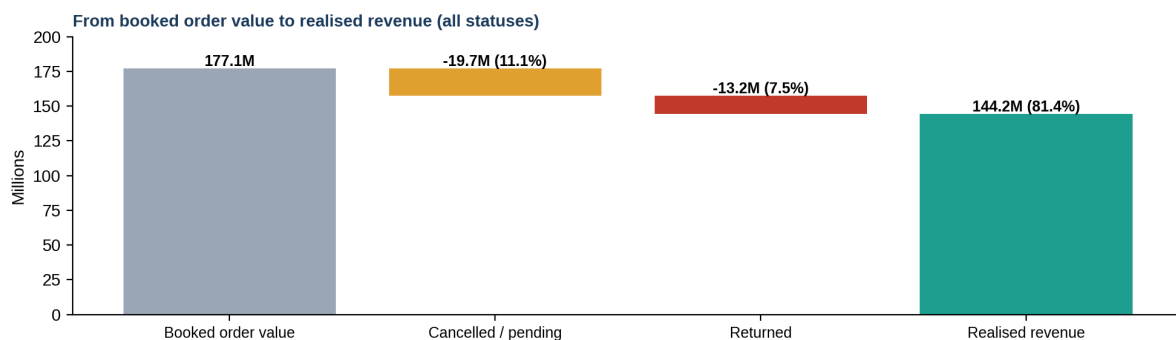
Check	Result	Action
Row counts, primary keys	All four tables match the source files; no duplicate keys	None
Orphan records	None in any join direction; 89 customers never ordered	LEFT JOIN used for customer summary
NULL values	One missing gender; other NULLs are structural (no return, no coupon, no campaign, no delivery for non-completed orders)	Gender set to Unknown in customers_clean
Invalid values	No zero or negative quantities, prices, sales or ratings	None
Outliers (IQR)	net_sales 6,435; profit 6,514; quantity 1,197; product price 96	Kept; they are genuine values
Formula integrity	net_sales = gross – discount + tax + shipping and profit = net_sales – product_cost – shipping hold for 100% of orders	None
Header vs item reconciliation	11,226 orders (8.13%) differ. Completed revenue is 156.8M (items) vs 144.2M (headers); profit 65.2M vs 59.8M	Headline KPIs use headers; difference disclosed
Payment status	11,306 Completed orders (10.0%) show payment Pending	Flagged; revenue may be overstated if these are unpaid
Repeated order-product pairs	87 pairs with different values	Kept; disclosed
Profit definition	Profit includes collected tax as income, so margins are slightly optimistic	Flagged; adjusted margin recommended
Loss-order counts	1,296 loss orders (0.94%) across all statuses; 1,229 (1.08%) among Completed	Completed figure used in this report

5. Headline Performance

KPI	Value	KPI	Value
Revenue	144.2M	Average order value	1,269.79
Profit	59.8M	Revenue per customer	5,827
Profit margin	41.44%	Orders per customer	4.6
Completed orders	113,559	Loss-making orders	1,229 (1.08% of Completed)
Active customers	24,748	Total realised loss	–191K (0.3% of profit)

Order status mix: Completed 82.2% (113,559), Returned 6.9% (9,462), Cancelled 6.1% (8,398), Pending 4.8% (6,697). **Delivery:** 14.9% of Completed orders (16,886) were delayed and 8.6% (9,757) arrived early.

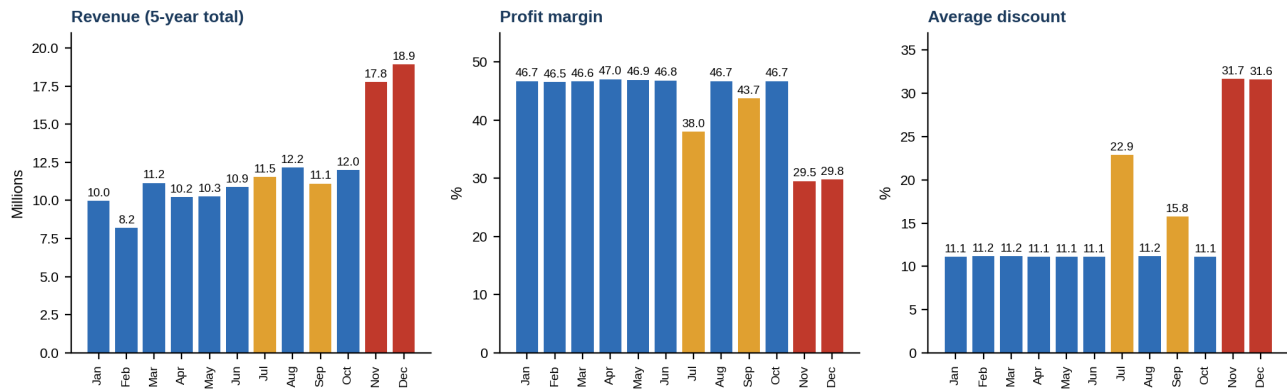
Revenue conversion. Booked value of 177.1M falls to 144.2M of realised revenue, a conversion of 81.4%: 19.7M (11.1%) is cancelled or pending and 13.2M (7.5%) is returned.



6. Findings

6.1 Seasonality: revenue peaks, profit does not

The monthly series shows a stable baseline (a typical month averages 2.15M of revenue, with February lowest at about 1.6M) and a sharp peak in November (3.55M on average) and December (3.78M), repeated almost identically in all five years. No sustained long-term growth is visible in the monthly trend. Profit barely moves: a typical month earns 0.98M, while November earns 1.05M and December 1.13M.



Month	Revenue (M)	Profit (M)	Margin	Avg discount	Orders	Loss orders
Jan	9.98	4.66	46.7%	11.1%	7,967	0
Feb	8.20	3.81	46.5%	11.2%	6,614	0
Mar	11.15	5.19	46.6%	11.2%	8,919	0
Apr	10.23	4.81	47.0%	11.1%	8,315	0
May	10.26	4.81	46.9%	11.1%	8,408	0
Jun	10.89	5.10	46.8%	11.1%	8,930	0
Jul	11.53	4.38	38.0%	22.9%	8,415	13
Aug	12.17	5.69	46.7%	11.2%	9,890	0
Sep	11.11	4.85	43.7%	15.8%	8,904	0
Oct	11.98	5.59	46.7%	11.1%	9,571	0
Nov	17.77	5.24	29.5%	31.7%	13,204	567
Dec	18.91	5.63	29.8%	31.6%	14,422	649
Total	144.20	59.76	41.4%		113,559	1,229

Completed orders, five-year totals by calendar month (header basis). Highlighted rows are the two peak months.

Hypothesis confirmed. Average discount is 11.1% in regular months, 15.8% in September, 22.9% in July and 31.6–31.7% in November–December, which matches the margin dips month by month. 98.9% of loss-making orders (1,216 of 1,229) fall in November–December and 1.1% (13) in July; no other month has a single loss-making order. The timing points to calendar-driven promotions (summer sale, back-to-school, Black Friday, holiday); the data does not name the campaigns, so the link to specific events remains an inference.

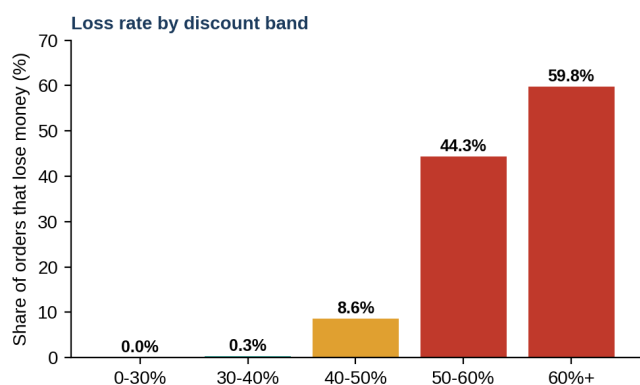
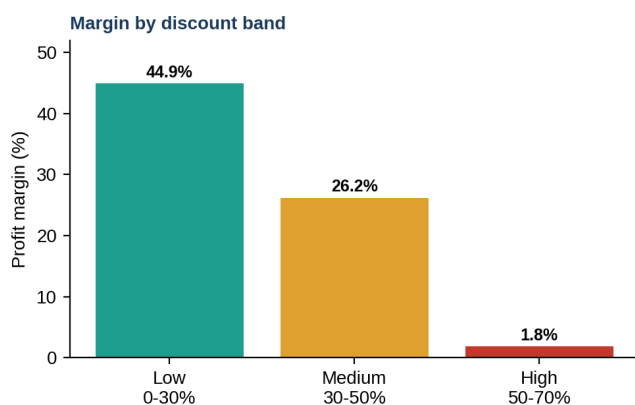
Implication

Against a typical month (average of the ten non-peak months), November–December added 15.2M of revenue over five years but only 1.1M of profit, an incremental margin of about 7% versus 45% in a typical month. Promotional volume is being bought at close to zero incremental margin. Part of this extra volume may be seasonal demand that would exist without the discounts, so the right test is a controlled promotion (recommendation 2).

6.2 Discount depth and profitability

Discount % is the strongest single driver of margin erosion in the data (correlation of discount amount with order-line margin is -0.52 ; shipping cost shows no relationship). Discount bands are assigned per order; revenue and profit are measured at order-line level (156.8M revenue basis).

Discount band (Completed)	Orders	Share of orders	Revenue	Margin	Loss-making orders	Loss rate
Low (0–30%)	96,266	84.8%	131.3M	44.9%	0	0.0%
Medium (30–50%)	15,663	13.8%	23.7M	26.2%	490	3.1%
High (50–70%)	1,630	1.4%	1.7M	1.8%	739	45.3%



A finer five-band SQL view shows the loss rate rising from 0% below 30%, to 0.3% at 30–40% (29 orders), 8.6% at 40–50% (461), 44.3% at 50–60% (675) and 59.8% above 60% (64). Losing orders average a 51.7% discount against 17.0% for profitable ones. Of the 1,229 loss-making orders, 1,200 (97.6%) are discounted more than 40%.

Sizing the leakage. If Medium and High orders had earned the Low-band margin of 44.9%, profit would be about 5.2M higher (4.4M from Medium, 0.7M from High). That equals 8.6% of headline profit (7.9% of the 65.2M line-basis profit it is calculated on). It is an upper bound, because some discounted orders would not exist without the discount.

Implication

The 30–50% band is where most margin is given away. The direct loss is minor; the margin loss is not.

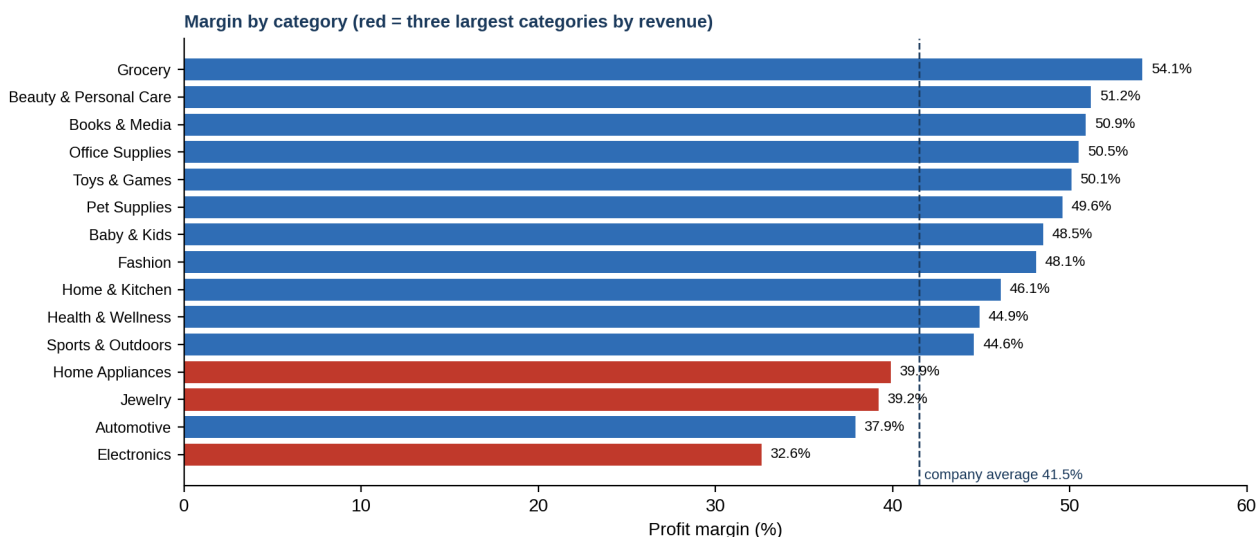
Methodological caution

Discount and margin are mechanically related, because a discount lowers net sales and therefore the margin denominator. The -0.52 correlation is a diagnostic signal, not a causal estimate. The 5.2M figure is an upper bound; a controlled promotion test (recommendation 2) is the stronger proof.

6.3 Category, subcategory and product mix

Category	Revenue (M)	Share of revenue	Profit (M)	Share of profit	Margin
Electronics	33.5	21.4%	10.9	16.7%	32.6%
Jewelry	20.8	13.3%	8.1	12.5%	39.2%
Home Appliances	20.3	12.9%	8.1	12.4%	39.9%
Automotive	13.4	8.5%	5.1	7.8%	37.9%
Sports & Outdoors	12.8	8.2%	5.7	8.8%	44.6%
Home & Kitchen	9.0	5.7%	4.1	6.3%	46.1%
Baby & Kids	8.0	5.1%	3.9	5.9%	48.5%
Health & Wellness	7.5	4.8%	3.4	5.2%	44.9%
Office Supplies	7.2	4.6%	3.7	5.6%	50.5%
Fashion	6.1	3.9%	2.9	4.5%	48.1%
Pet Supplies	5.2	3.3%	2.6	4.0%	49.6%
Toys & Games	4.5	2.9%	2.2	3.4%	50.1%
Beauty & Personal Care	3.9	2.5%	2.0	3.0%	51.2%
Grocery	2.4	1.6%	1.3	2.0%	54.1%
Books & Media	2.3	1.5%	1.2	1.8%	50.9%

Line basis (156.8M revenue). Categories sorted by revenue.



Margin falls as ticket size rises: the four highest-value categories (Electronics, Automotive, Jewelry, Home Appliances) all earn below 40%, while the low-priced categories (Grocery, Books, Beauty, Toys, Pet Supplies, Office Supplies, Fashion) earn 48% or more. Within Electronics, the largest subcategories are Smartphones (6.3M), TVs (5.7M), Smart Watches (5.2M) and Tablets (3.7M); the lowest-margin subcategories are Tablets (30.4%), Headphones (31.2%), TVs (31.8%) and Smart Watches (31.9%). The best margins are in Grocery Snacks (60.7%), Fashion Bags (56.7%) and Deodorants (56.2%). Bags is notable because it combines a high margin with meaningful revenue (1.3M) and is a candidate for promotion.

At product level, none of the 1,175 products is loss-making in aggregate, so losses are a transaction and discount problem, not a product problem. Revenue is broadly distributed: 45.3% of products are needed to reach 80% of revenue. A median-split matrix classifies 550 products as high revenue / high profit, 549 as low / low, and 38 each as high revenue / low profit and low revenue / high profit. The 38 high-revenue, low-profit products are the first candidates for pricing and discount review. The top profit contributor is PROD-000996 (303K profit, 52.6% margin).

Implication

Each additional point of Electronics margin is worth about 0.33M over the period, which makes Electronics the largest single margin lever by mix.

6.4 Customers and segments

Segment	Customers	Orders per customer	Revenue per customer	Share of revenue	Margin
Consumer	13,508	4.6	6,387	55.0%	42.7%
Premium	6,230	4.6	6,263	24.9%	39.4%
VIP	2,516	4.6	6,280	10.1%	39.5%
Business	2,494	4.5	6,297	10.0%	42.6%

Line basis, so revenue per customer (about 6.3K) is higher than the headline 5,827 on a header basis.

Segments are almost identical in spend and frequency, and Premium and VIP earn about 3 points less margin, which is consistent with heavier discounting (not tested directly). VIP customers are not more valuable than Consumer customers in this data. Customer concentration is negligible: the largest customer contributes 33.3K, about 0.02% of revenue. The few loss-making customers are small (the worst is -1,070 on 7.7K revenue). 252 customers have no Completed order, including 89 who never placed any order.

Implication

The existing segmentation cannot be used to target retention or pricing; behavioural segmentation (RFM) is needed.

6.5 Regions and shipping

Region	Revenue (M)	Share	Margin	Shipping cost as % of revenue
South	49.3	31.5%	40.6%	1.81%
Central	32.8	20.9%	40.5%	1.82%
West	30.8	19.7%	43.5%	1.60%
East	24.0	15.3%	40.0%	1.91%
North	19.8	12.7%	44.5%	1.55%

Regional revenue follows the size of the customer base (revenue per customer is 6.1–6.6K everywhere). The margin spread is 4.5 points: South, Central and East are below the company average and North and West above it. Shipping is only 1.5–1.9% of revenue and uncorrelated with profit, so it is a minor lever.

6.6 Revenue leakage

Only 81.4% of booked order value becomes revenue. Returned orders alone represent 13.2M, equal to 9.2% of realised revenue. Pending orders may still convert, so the 19.7M of cancelled and pending value is not all lost; the split between the two has not been valued. Return reasons, cancellation causes and the 11,306 Completed orders with Pending payment have not yet been analysed and represent the largest unexplored opportunity.

7. Recommendations

#	Action	Evidence	Estimated impact	Effort
1	Hard discount cap at 40%; manager approval for 30–40%; automatic block when expected margin falls below a floor	0% loss below 30%; 8.6% at 40–50%; 44–60% above 50%. 1,200 of 1,229 loss orders are above 40%	Removes most of the 191K direct loss; capturing 25–50% of the 5.2M leakage equals 1.3–2.6M over five years	Low
2	Redesign Nov–Dec promotions: targeted bundles and loyalty offers instead of blanket discounts; test on one category first	Margin 29.5–29.8% in peak vs 46–47% normally; average discount 31.7% vs 11.1%	Peak months carry 36.7M of revenue: each margin point is worth 0.37M over five years (about 73K a year); recovering half the gap (about 8.5 points) is worth about 3.1M	Medium
3	Protect Electronics margin: cap discount depth, bundle accessories, review the 38 high-revenue / low-profit products	21.4% of revenue at a 32.6% margin	+1 point of margin is about 0.33M over the period	Medium
4	Rebuild segmentation with RFM and CAC	All four segments show identical spend and frequency	Enables targeted retention and pricing	Medium
5	Investigate cancellations, returns and Completed orders with Pending payment	32.9M of booked value (18.6%) not realised	A 10% reduction in returns protects about 1.3M of booked value (about 0.55M of profit at the 41% margin)	Medium
6	Resolve data issues: header vs item gap (8.7%), tax counted in profit, payment-status inconsistency	Section 4	Improves trust in every KPI	Medium

Impacts are five-year scenarios. Recommendations 1 and 2 act on the same discounted orders (most of the peak-season discounting sits above 30%), so only one of the two estimates should be counted in a business case.

Recommended implementation sequence

Phase	Focus	Actions
1 Control	Guardrails and definitions	Discount cap and approval flow (rec. 1); KPI definitions and data fixes (rec. 6)
2 Optimise	Peak season and pricing	Targeted Nov–Dec bundles (rec. 2); Electronics pricing and bundling (rec. 3)
3 Personalise	Customer value	RFM and CAC-to-profit segmentation; retention offers (rec. 4)
4 Validate	Leakage root causes	Returns, cancellations and Pending-payment analysis (rec. 5)

How to measure success after launch: share of orders discounted above 30%; number of loss-making orders; November–December margin; Electronics margin; return rate; share of Completed orders with Pending payment.

8. Limitations and Assumptions

- Headline KPIs use order headers; category, product, segment, region and line-level discount analysis use order items. Totals differ by about 8.7% (profit by 9.0%), so the two views must not be added together.
- Currencies are summed without conversion because the data is on a common scale.
- Profit includes tax collected, so true margins are somewhat lower than reported.
- Delivery data exists only for Completed orders, so the effect of late delivery on returns cannot be measured.
- 11,306 Completed orders show Pending payment; revenue could be overstated.
- The dataset appears to be synthetic, and some patterns (identical spend across segments, uniform behaviour) may be artefacts of data generation. Conclusions should be validated on real business data.
- Seasonal figures are read from the monthly summary and rounded. The link between peak-season discounting and margin is strong and consistent across months, but it is an association; named campaigns are not identified in the analysis.
- The statement of no long-term growth is based on the monthly trend; yearly revenue and year-over-year change are shown on the Power BI Executive Overview page.
- Impact estimates are scenarios, not forecasts.

9. Next Steps

1. Add a 6–12 month revenue and profit forecast (seasonal model) from the monthly summary view.
2. Add RFM segmentation and a CAC-to-profit view using customer_acquisition_cost.
3. Analyse return reasons, cancellations, sales channel, marketing channel and coupon/campaign profitability, which are present in the data but not yet used. Campaign and coupon data can also name the promotions behind the Nov–Dec discounts.
4. Value the 11,306 Completed orders with Pending payment and the cancelled vs pending split of the 19.7M.
5. Publish the Power BI dashboard with a data notes page and a one-page executive summary.

10. How This Analysis Applies to a Real Business

The same method can be run on a company's own sales data in a few steps: audit the data and reconcile it to the finance totals; build a clean SQL layer; find where margin is lost by discount, product, customer, region and season; and deliver an interactive Power BI dashboard with a short action list. The questions answered here (where profit leaks, which discounts pay back, which customers and products matter) are relevant to any business that sells at varying prices.

Appendix: Project Files

File	Purpose
Sales_Profitability_Intelligence.sql	Data audit, reconciliation, clean views
Sales_Profitability_Intelligence.ipynb	Python EDA, business analysis and exported analysis CSVs
SALES_PROFITABILITY_INTELLIGENCE.pbix	Four-page interactive dashboard: Executive Overview; Product & Category; Customers & Regions; Discount & Loss
Sales_Profitability_Intelligence_Analytical_Report_FINAL.pdf	This report: findings, data quality, recommendations and limitations

Prepared by Nijat Aliyev, Data Analyst, Warsaw, Poland. Stack: SQL Server · Python (pandas, matplotlib, seaborn) · Power BI. Analysis period: January 2021 – December 2025.